PACKET 1 • SOFTWARE DEVELOPMENT

1 Use AI the Way You Normally Do

RefView Window/Level Display Transform

Workshop sandbox RefView is fictional, created solely for this workshop, and not intended for clinical use.

Challenge

Design and implement a reusable component that converts CT-like pixel values into 8-bit grayscale display values using a window center and window width.

- Input: pixel value, window center, and window width.
- Output: an integer display value from 0 through 255.
- Values below the window appear black; values above it appear white; values inside are mapped proportionally.

Deliverable

- Interface definition
- Implementation
- Automated unit tests
- Documented assumptions or unresolved questions

Constraints

- The component will be called once for every pixel in an image.
- The result must always be an integer from 0 through 255.
- The same inputs must always produce the same output.
- Invalid inputs must be handled predictably.

Acceptance criteria — fixed throughout the workshop

1. Correctly maps values below, inside, and above the specified window.
2. Always returns an integer in the range 0–255.
3. Produces deterministic results.
4. Defines predictable behavior for invalid window parameters.
5. Includes tests covering normal cases, boundaries, and invalid inputs.

Your task Use AI the way you normally do to design, implement, and test the component.

PACKET 2 • RESEARCH SCIENTISTS

1 Use AI the Way You Normally Do

RefView SmoothCT Experimental Evaluation

Workshop sandbox RefView and SmoothCT are fictional and do not represent an actual product or algorithm.

Scenario and hypothesis

The RefView research team claims that SmoothCT reduces image noise while preserving image quality.

Hypothesis SmoothCT reduces image noise by at least 30% without materially degrading spatial resolution or CT-number accuracy.

Deliverable

Develop an experimental plan describing measurements, experimental conditions, controls, observations or repetitions, analysis approach, uncertainty, and decision criteria.

Acceptance criteria — fixed throughout the workshop

1. Quantitatively evaluates noise, spatial resolution, CT-number bias.
2. Includes an appropriate matched control.
3. Addresses repeatability and measurement uncertainty.
4. Identifies important potential confounding variables.
5. Defines objective criteria for deciding whether the hypothesis is supported.

Success thresholds

Measure	Required result
Image noise	Decrease of at least 30%
Spatial resolution	Degradation no greater than 5%
CT-number bias	Absolute introduced bias no greater than 3 HU

Your task Use AI the way you normally do to develop the experimental plan.

PACKET 3 • PRODUCT DEVELOPMENT

1 Use AI the Way You Normally Do

RefView Auto-Window Capability

Workshop sandbox RefView and Auto-Window are fictional, not an actual product capability, and not intended for clinical use.

Scenario

RefView currently opens every image with the same default window. The proposed Auto-Window capability analyzes an image on opening and selects an initial window center and width. Users remain free to adjust the result manually.

Challenge

Develop product/system requirements, technical risks, and a verification plan for Auto-Window.

Basic requirements

- Automatically select an initial window center and width.
- Operate without user interaction.
- Provide a useful initial display for typical images.
- Allow the user to override the result.
- Behave predictably when an appropriate window cannot be determined.

Acceptance criteria — fixed throughout the workshop

1. Objectively testable functional requirements.
2. Measurable performance requirements.
3. Behavior for abnormal or unsupported inputs.
4. Significant technical risks and mitigations.
5. Verification evidence for every requirement.
6. Objective pass/fail criteria.
7. Representative normal and edge-case images.

Your task Use AI the way you normally do to develop the requirements, risks, and verification plan.